

Eli Lewis

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EDUCATION

University of Illinois at Urbana-Champaign

August 2023 – May 2027

Bachelor of Science in Aerospace Engineering — M.S. Planned, Fall 2027

GPA: 4.00/4.00

Relevant Coursework: Aerospace Control Systems, Aircraft Flight Mechanics, Spacecraft Rendezvous and Proximity Operations, Orbital Mechanics, Numerical Methods, Autonomous Systems Lab, Aerospace Dynamical Systems, Aerospace Systems Design

Academic Honors: State Farm National Merit Scholarship, Engineering Visionary Scholarship, Dean's List (5 Semesters)

EXPERIENCE

Draper

January 2026 – June 2026

Guidance, Navigation, and Control (GNC) Engineer Co-op, Hypersonics and Advanced Weapons

Cambridge, MA

- Developed Draper's in-house 6DOF GNC Simulink framework supporting air and space vehicle programs, architecting modular vehicle dynamics, actuator, sensor, and GNC components for rapid vehicle integration and simulation.
- Formulated nonlinear EKF and UKF state estimators, re-deriving measurement Jacobians to incorporate additional sensor measurements, improve state observability, and increase mission success rate by 43%.
- Designed and validated nonlinear flight control algorithms for an under-actuated, under-sensed air vehicle, improving mission success rate by 26% across nominal conditions.
- Architected Monte Carlo simulations to characterize vehicle performance across mission, environmental, and model uncertainties, informing system-level trades and component requirements.
- Performed trajectory generation and optimization for early-stage vehicle definition, implementing preliminary guidance and control architectures and mission constraints to evaluate vehicle performance and feasibility.

Machine Learning Reliability Research – Dept. of Aerospace Engineering

January 2025 – December 2025

Undergraduate Research Assistant

Champaign, IL

- Built and trained neural networks using PyTorch to output both the model target regression predictions and a reliability metric for detecting out-of-distribution data and anomalous predictions.
- Integrated checksums with uncertainty quantification to improve model reliability in critical scientific and engineering applications.
- Developed a composite loss function combining prediction accuracy with checksum penalty and reward terms.
- Achieved 99.7% AUROC on a chemical combustion dataset, cutting undetected out-of-distribution samples from 97.7% to 0.1%.

City of Bloomington Engineering Department

May 2025 – August 2025

Engineering Intern

Bloomington, IL

- Created the 2027 road resurfacing plans for the City of Bloomington, detailing scope, material requirements, and utility coordination.
- Built a city-wide crash database to centralize historical crash data for analysis and engineering intervention to support intersection safety improvements.

ORGANIZATIONS & LEADERSHIP

Liquid Rocketry at Illinois

August 2024 – May 2025

Controls Sub-Team Lead

Champaign, IL

- Directed an interdisciplinary GNC software team in C++ and Simulink to control roll rate for a model rocket launch.
- Automated ANSYS CFD workflow with Python to model nonlinear fin deflection behavior, generating aerodynamic coefficient lookup tables to parameterize vehicle roll-control authority.
- Sized actuated fin deflector geometry and deflection limits ($\pm 15^\circ$), maximizing roll-control authority and maneuverability while maintaining passive longitudinal stability across the flight envelope.

Illinois Electric Motorsport (Formula SAE)

August 2023 – May 2024

Aerodynamics Team Member

Champaign, IL

- Created meshes of aerodynamic structures in Star-CCM+; applied refinement tools to improve accuracy of mesh and CFD simulations.
- Modeled airfoil elements and mounting structures in PTC CREO, utilizing JavaFoil to guide design choices.
- Performed CFD sweeps in Star-CCM+ of diffuser designs; extrapolated results to determine optimal angle and sizing.

SKILLS

Programming: Python, C++, MATLAB

Tools and Frameworks: Simulink, PyTorch, Linux, ANSYS, Star-CCM+, NX, PTC Creo

Technical Skills: GNC, EKF/UKF, 6DOF Dynamics, V&V, MBSE, Numerical Methods, Simulation and Modeling